Folha 4

Exercício 4.2

a

```
ArcSin[Sin[5 Pi / 4]]
```

$-\frac{\pi}{4}$

b

```
Sin[ArcSin[-1 / 2]]
```

$-\frac{1}{2}$

c

```
Sin[ArcSin@1 + Pi]
```

-1

d

```
ArcSin[Sin[-Pi / 6]]
```

$-\frac{\pi}{6}$

e

```
ArcSin[Sin[23 Pi / 6]]
```

$-\frac{\pi}{6}$

f

```
Cos[ArcCos[1 / 8]]
```

$\frac{1}{8}$

g

```
ArcCos[Cos[-Pi / 3]]
```

$\frac{\pi}{3}$

h

```
ArcTan[Tan[9 Pi / 4]]
```

$\frac{\pi}{4}$

i

```
ArcTan[Tan[Pi]]
```

0

j

```
Tan[ArcTan[-1]]
```

-1

k

```
Tan[ArcCot[3]]
```

$\frac{1}{3}$

l

```
ArcTan[Cot[Pi / 5]]
```

$\frac{3\pi}{10}$

Exercício 4.4

a

```
ArcSin[-√2/2]
```

$-\frac{\pi}{4}$

b

```
Cot[ArcSin[-4 / 5]]
```

$-\frac{3}{4}$

c

```
Cos[ArcSin[1 / 2] - ArcCos[3 / 5]] // TrigExpand
```

$\frac{2}{5}+\frac{3\sqrt{3}}{10}$

d

```
Sin[Pi - ArcSin@1]
```

1

e

```
Sin[Pi / 2 + ArcCos[Sqrt@3 / 2]]
```

$\frac{\sqrt{3}}{2}$

f

```
Sin[ArcTan[-1]]
```

$-\frac{1}{\sqrt{2}}$

g

```
Sin[ArcCos[Sqrt@2 / 2]]
```

$\frac{1}{\sqrt{2}}$

h

```
Cos[-2 ArcSin[-3 / 5]] // TrigExpand
```

$\frac{7}{25}$

i

```
Tan[-ArcSin[Sqrt@2 / 2]]
```

-1

j

```
ArcTan[-2 + Tan[5 Pi / 4]]
```

$-\frac{\pi}{4}$

k

```
ArcSin[Sin[Pi / 2]] + 2 ArcCos[-Sqrt@2 / 2]
```

2π

l

```
Cos[1 / 2 ArcCos[1 / 3]] ^2 - Sin[1 / 2 ArcCos[1 / 3]] ^2 // Simplify
```

$\frac{1}{3}$

m

```
Tan[ArcSin[3 / 5]] ^2 - Cot[ArcCos[4 / 5]] ^2
```

$-\frac{175}{144}$

Exercício 4.5

```
g[x_] = Pi / 3 + 2 ArcSin[1 / x];
```

a

$\frac{4\pi}{3}$

b

Dg = $(-\infty,-1] \cup [1,+\infty)$ CDg = $\left[\frac{-2\pi}{3}, \frac{4\pi}{3}\right] \setminus \left\{\frac{\pi}{3}\right\}$

c

$(-\infty,-1] \cup [2,+\infty)$

d

$g^{-1} : \left[\frac{-2\pi}{3}, \frac{4\pi}{3}\right] \setminus \left\{\frac{\pi}{3}\right\} \longrightarrow (-\infty,-1] \cup [1,+\infty)$, $g^{-1}(x) = \frac{1}{\operatorname{Sin} \frac{3x-\pi}{6}}$

Exercício 4.6

a

Descontínua (em -1).

b

$\left[\frac{-\pi}{2}, \frac{\pi}{2}\right]$

c

Zero e não existe, respetivamente.

Exercício 4.7

a

$\frac{2}{\pi}$

b

Ambos zero.

Exercício 4.8

a

```
x /. Solve[Exp[x] == Exp[1 - x], x, Reals]
```

$\left\{\frac{1}{2}\right\}$

b

```
x /. Solve[Exp[2 x] + 2 Exp[x] - 3 == 0, x, Reals]
```

$\{0\}$

c

```
x /. Solve[Exp[3 x] - 2 Exp[-x] == 0, x, Reals]
```

$\left\{\frac{\operatorname{Log}[2]}{4}\right\}$

d

```
x /. Solve[Log[x^2 - 1] + 2 Log[2] == Log[4 x - 1], x, Reals]
```

$\left\{\frac{3}{2}\right\}$

Created with the Wolfram Language